MCPA / SiteTrace

MCPA

Construction Equipment Accountability and Inventory Management System

A centralized platform supporting organized equipment, project, and inventory operations.

Equipment **Movement** Projects Consumables

OPERATIONAL NEED

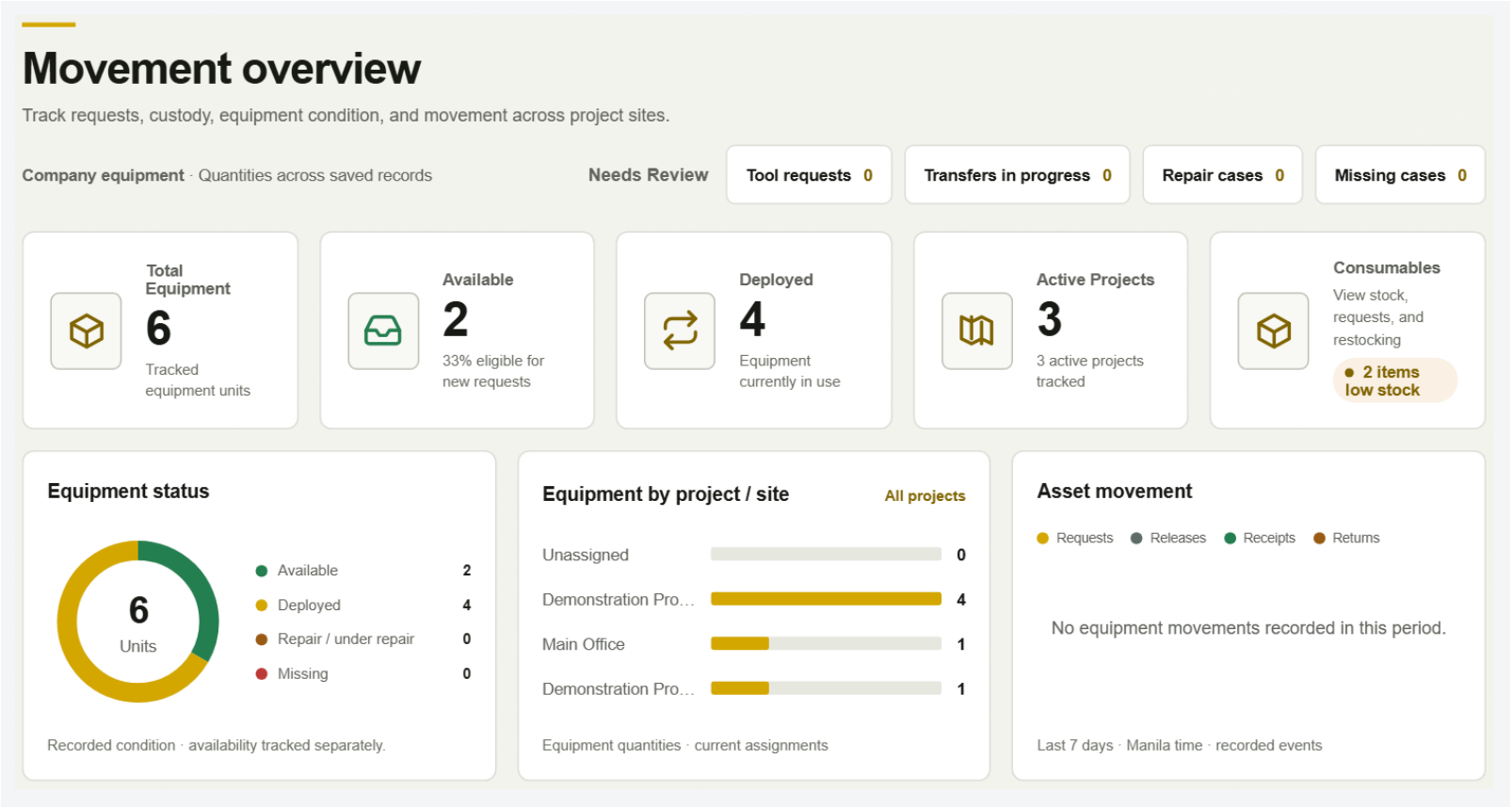
The Challenge of Equipment Accountability

OPERATIONAL CHALLENGE	IMPLEMENTED CAPABILITY	SUPPORTED VALUE
Fragmented equipment records	**Shared equipment inventory**	One reference for asset information
Unclear responsibility	**Named holder + inspected receipt**	Visible recorded custody
Hard-to-trace movements	**Linked requests, transfers and activity**	Reviewable movement sequence
Limited project visibility	**Equipment linked to project records**	Visible deployment context
Consumable stock uncertainty	**SKU balances + low-stock thresholds**	Structured stock review

These are supported capabilities. Time savings, cost reductions and business impact have not been measured.

Centralized Operational Visibility

Inventory & availability | Project allocation | Review queues



Admin

Company overview; request review and equipment release.

Engineer · Architect

Own custody and assigned projects; requests, transfers and receipt.

Tool Handler

Inventory monitoring; dispatch of available, unheld equipment.

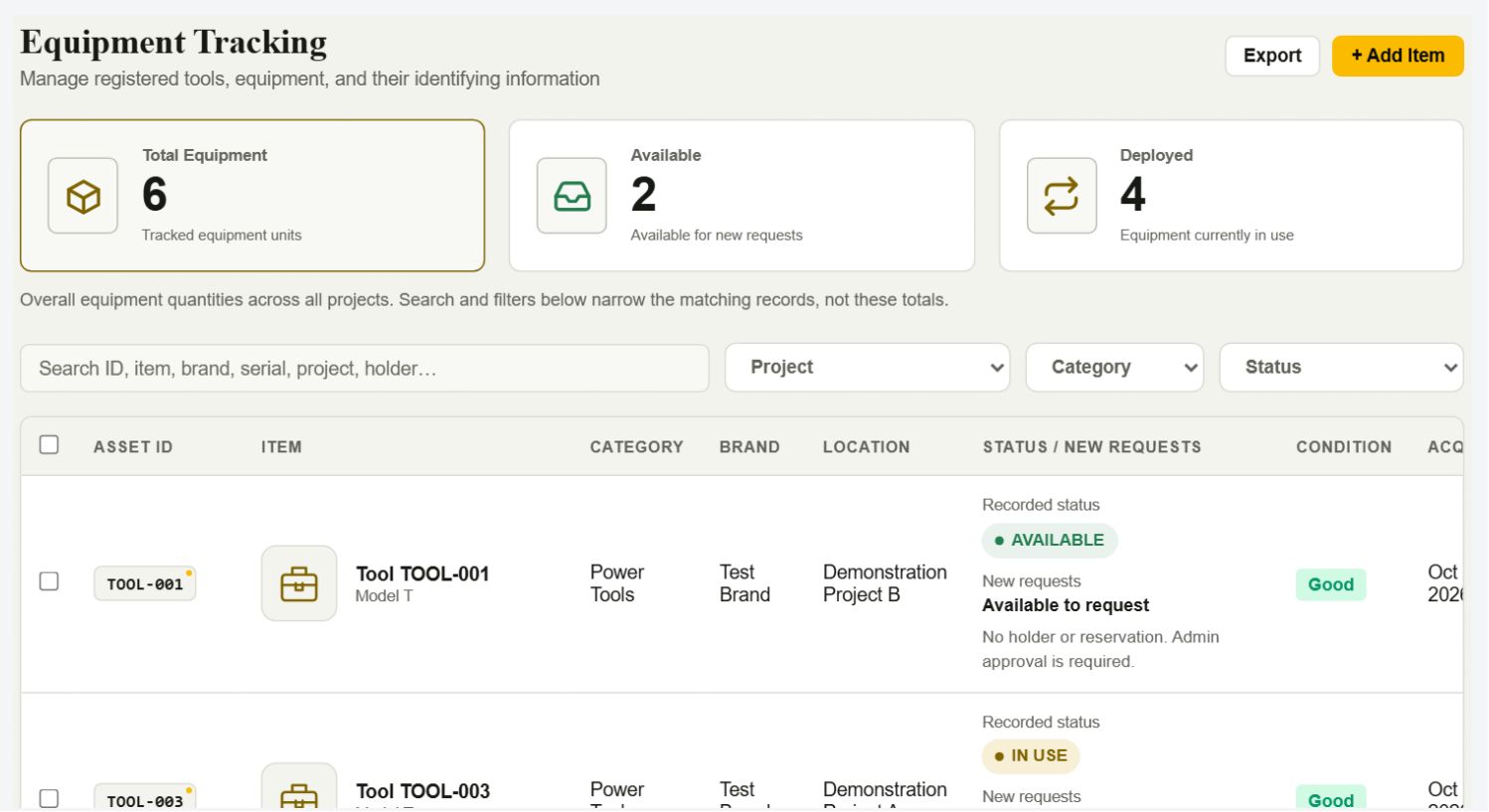
Secretary

Consumables and purchasing; equipment routes are restricted.

ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

Knowing What Equipment Exists and Who Is Responsible

EQUIPMENT REGISTER



RECORDED CUSTODY



Identify the asset

Asset ID, name and registration specifications.

Review responsibility

Saved holder and project; status and request availability.

ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

CORE WORKFLOW

From Equipment Request to Accountable Handover

CUSTODY CHANGES HERE

01 REQUEST

Engineer / Architect

request: pending

Select eligible equipment and an assigned project. Custody stays unchanged.

02 REVIEW

Admin

request: approved

Approve to reserve tools. Reject with a recorded reason. Rejected requests stop here.

03 RELEASE

Admin

request: released
transfer: pending

Create the named handover and retain a source snapshot. Custody stays unchanged.

04 INSPECT + RECEIVE

Named Engineer / Architect

request + transfer: received

Record every item's condition and confirm physical testing. Accepted items change custody.

TRANSFER TO ANOTHER PROJECT

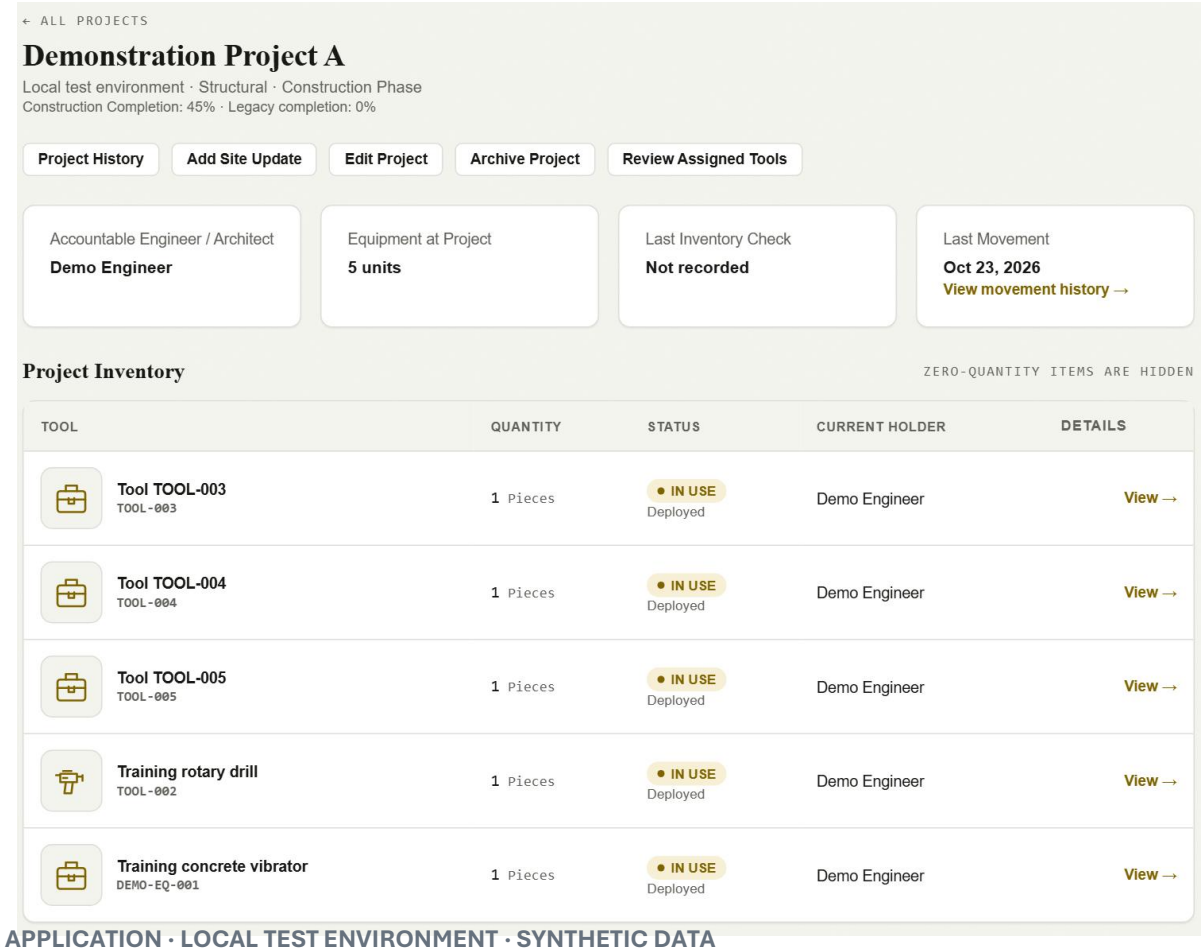
Engineer / Architect sends → named receiver inspects.

RETURN TO CONFIGURED OFFICE

Engineer / Architect; good / damaged clear custody.

Exceptions: accepted damage → repair case; missing item → prior holder and last recorded site retained.

Connecting Equipment to Construction Projects



ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

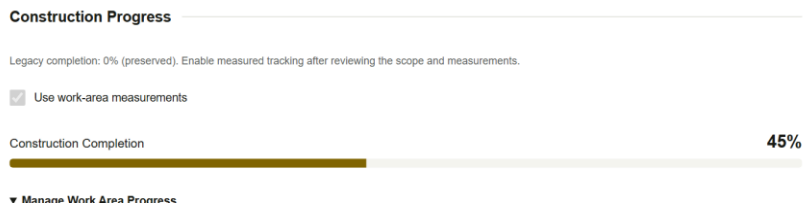
Assigned responsibility

Project information links to an Engineer / Architect profile.

Deployment visibility

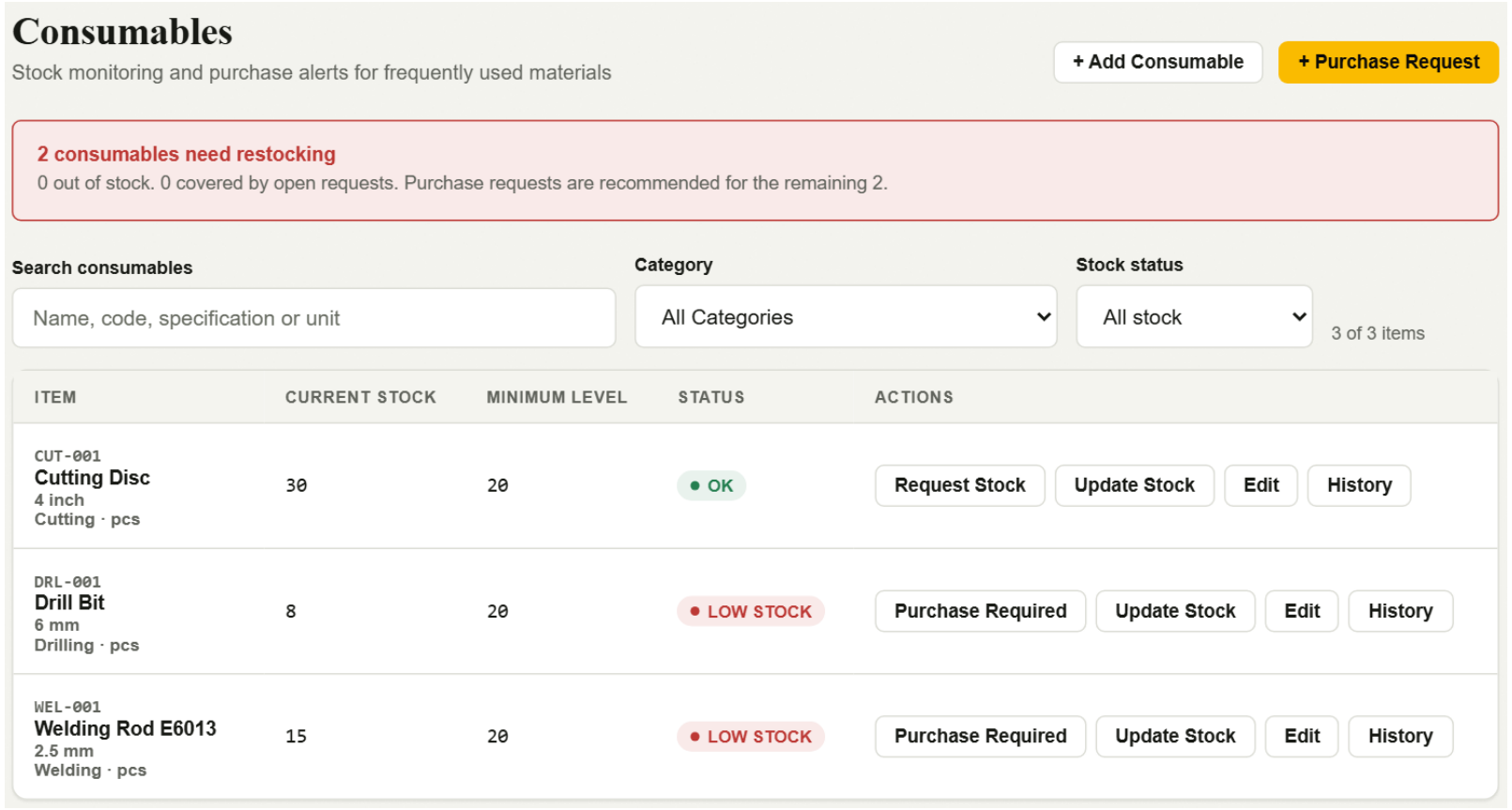
Review linked equipment, saved holders and recorded statuses.

WEIGHTED COMPLETION



Calculated from configured work-area weights. Locally verified upgrade; production installation unconfirmed.

Managing Materials and Consumable Stock



Organized catalog

Item codes, categories, variants and specifications.

Stock visibility

Saved quantity, minimum level and low-stock indicators.

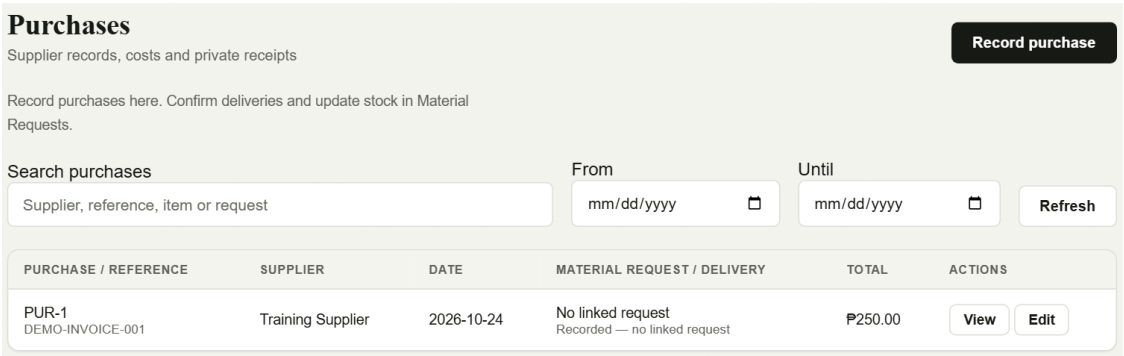
Controlled movements

Restock, usage and adjustments; request receipt updates stock.

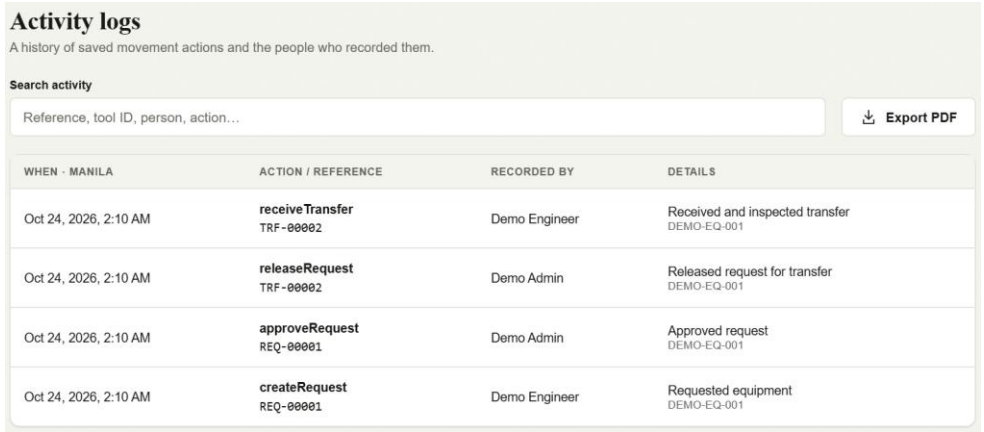
ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

Records That Support Operational Traceability

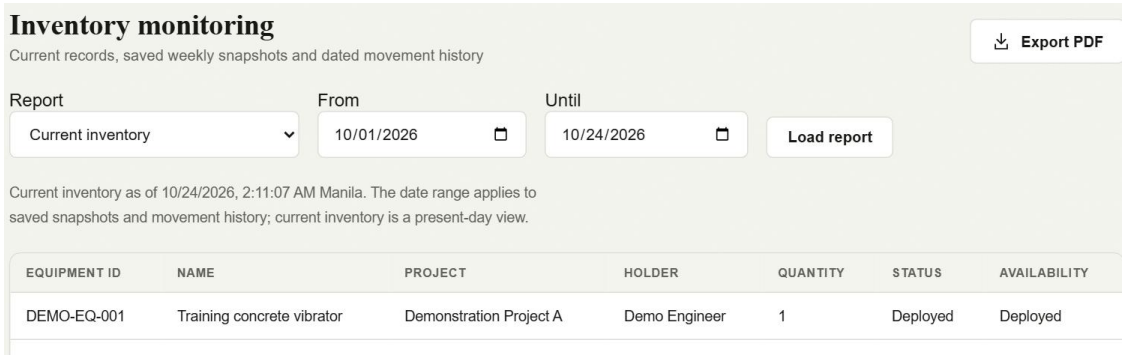
PURCHASING LEDGER



MOVEMENT ACTIVITY



CURRENT INVENTORY REPORT



Retained evidence

Purchase history and private receipt files
Movement actors, timestamps and references
Server-side role and record checks

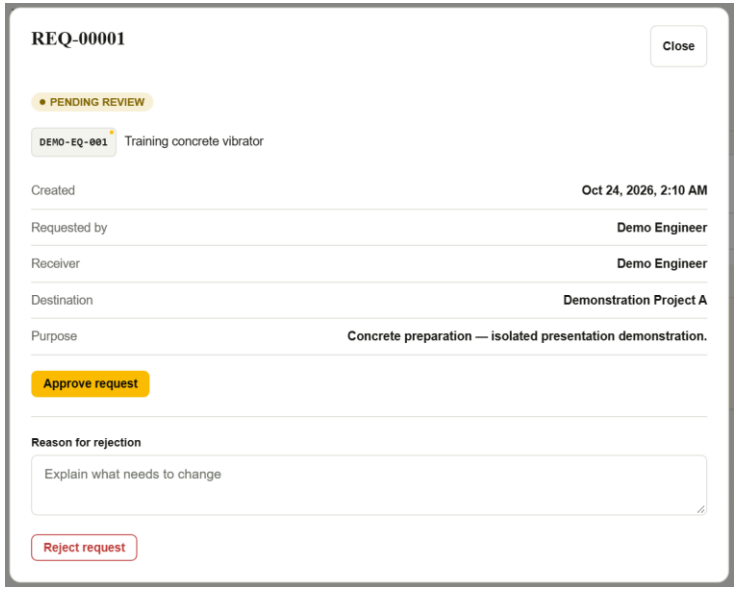
ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

A Practical Equipment Request Scenario

DEMO-EQ-001 · Training concrete vibrator

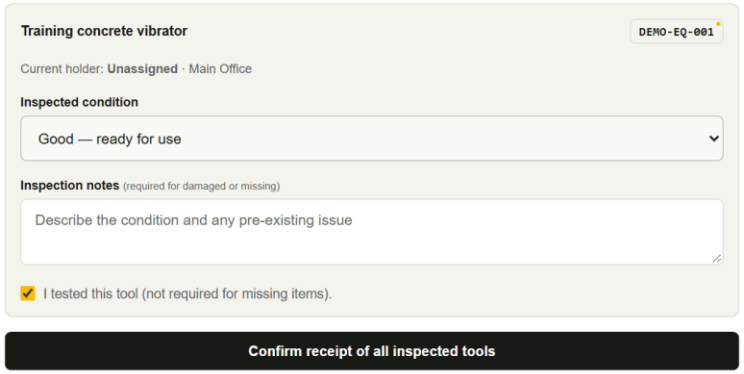
Executed in the local test application; no production transaction.

01 REQUEST + REVIEW



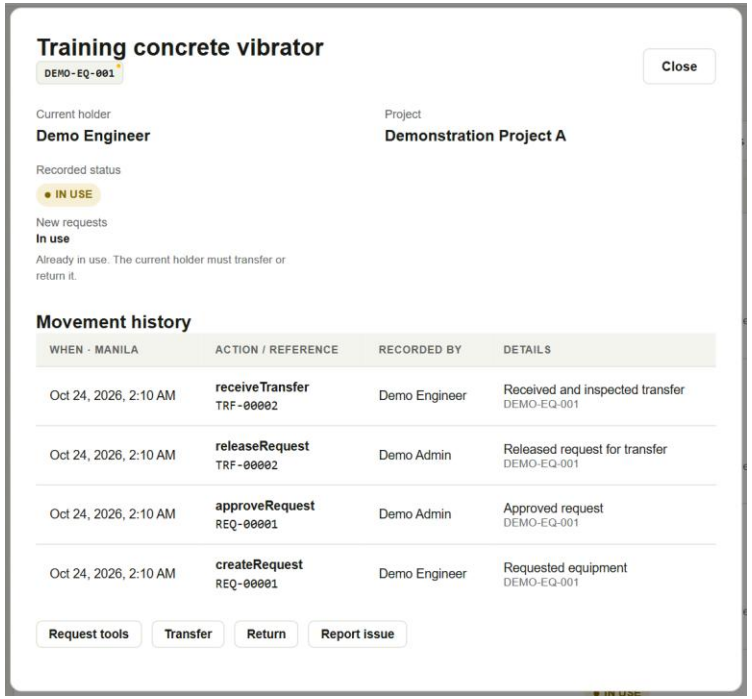
pending → approved
Holder remains unassigned.

02 RELEASE + INSPECTION



released → received
Test + condition required.

03 UPDATED CUSTODY



ACTUAL APPLICATION · LOCAL TEST ENVIRONMENT · SYNTHETIC DATA

Building Accountability Into Construction Operations

- Centralized equipment information
- Clearer recorded responsibility
- **Traceable saved equipment movements**
- Visible project deployments
- Organized consumable stock records

MCPA / SiteTrace — Every Asset Accounted For.
Every Movement Traceable.

SYSTEM VISION · supported capabilities, not measured business outcomes